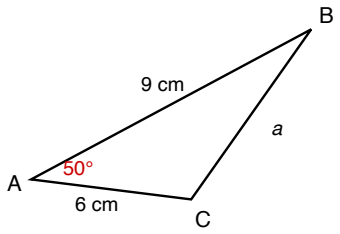
Shape and Space 10 — Graphs of sin, cos and tan · Sine rule · Cosine rule · Area of a triangle

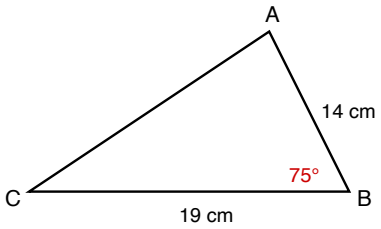
Calculator allowed. Marks in brackets. ★ = harder question. Lengths to 3 s.f., angles to 1 d.p. Diagrams not drawn accurately.

1 **a** Sketch the graph of $y = \cos\theta$ for $0° \le \theta \le 360°$. **b** Use your graph to find all solutions of $\cos\theta = \dfrac{1}{2}$ in that range. [7]

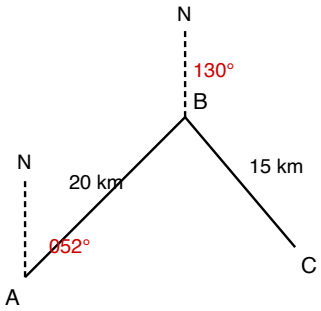
2 In triangle ABC, angle A $= 50°$, AB $= 9$ cm and AC $= 6$ cm. Find **a** the length a = BC **b** angle B. [6]



3 In triangle ABC, angle B $= 75°$, AB $= 14$ cm and BC $= 19$ cm. Find **a** angle C **b** the area of the triangle. [6]



4 Town B is 20 km from town A on a bearing of 052°. Town C is 15 km from B on a bearing of 130°. **a** Calculate the direct distance AC. **b** Calculate the bearing of C from A. [6]



Total: 25 marks

Check: sine graph starts at 0, cosine at 1; both repeat every 360° · second solution: sin → $180° - \theta$, cos → $360° - \theta$ · two sides + included angle, or three sides → cosine rule; otherwise sine rule · area $= \frac{1}{2}ab\sin C$ · bearings: find the angle inside the triangle first, using north lines and $180°$ back-bearings.